

CELLCHECK · DESIGN DECISIONS REGISTER

CellCheck design decisions register

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DOCUMENT CONTROL

Document ID	CCK-DEC-001	Version	0.6
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-03	Amish Chadha	Amish accepted the cost overrun against the value-engineering target on 2026-10-03; row added to decisions made; value engineering section updated
0.5	2026-10-02	Amish Chadha	Estimated cost updated to \$177.60 (\$2.60 over the target) after the fan finger guards and silicone were priced; open decision 1 added for the overshoot
0.4	2026-10-02	Amish Chadha	Design for construction (CCK-DDR-003, P1 to P12) accepted by Amish on 2026-10-02; the 2026-10-01 row no longer says open for review
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations of open items 1 to 11 on 2026-10-02; all moved to decisions made; value-engineering savings line updated
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target

0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note and decision records, the build plan work and every decision made so far
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CellCheck design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. Open decision 1 (the USD 2.60 overshoot) was decided on 2026-10-03.

■ To confirm when parts are bought

Table 2. Items to check against the parts before building.

#	What to confirm	Why it matters	Source
1	The steel tray is plain steel, about 460 x 300 x 45 mm with 1.5 mm walls, and its flat floor is at least 457 x 297 mm inside	The liner, plate and containment estimate assume it	CCK-DDR-003, P1
2	The rubber feet have M4 studs about 8 mm long	The stud must pass the 1.5 mm floor and engage the standoff	CCK-DDR-003, P1
3	The M4 rivet nuts grip 1.5 mm sheet and their heads are under 1 mm thick	The guard flange sits round the head	CCK-DDR-003, P4
4	The cell holders have two M3 fixing holes in their base and take 18650 and 21700 cells up to 70 mm	Holder screws and the 84 mm holder length	CCK-DDR-003, P3
5	The charger modules, current monitor breakouts, controller and converter have M3 mounting holes (or are fixed to a small perfboard that has them)	Each board stands on two nylon standoffs	CCK-DDR-003, P5
6	The 2.8 in display module is about 86 x 50 mm with four corner holes	Display stand size	CCK-DDR-003, P6
7	The heatsink spine is 6 mm thick, 50 mm tall, and has a fin gap at 20 mm from the left end and every 40 mm after it	MOSFET screws go between fins; plate and shroud screws are tapped into the spine	CCK-DDR-003, P7
8	The fans have a 50 mm square screw pattern	Shroud holes	CCK-DDR-003, P8
9	The DC jack, fuse holder and rocker switch cut-out sizes against the inlet housing (11.2 mm, 12 mm, 12 x 16 mm)	Inlet housing holes	CCK-DDR-003, P10
10	A sourced figure for the heat released by one 21700 cell in thermal runaway, to replace the assumed 100 kJ	Containment estimate before any TRL 4 work	CCK-CAL-001 section 7

■ Value engineering

Value-engineering target: USD 175.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 177.60 (USD 2.60 over the target).

Amish accepted this overrun on 2026-10-03: the estimated cost of USD 177.60 against the USD 175 target (USD 2.60 over; open decision 1, option a). Amish: "Cost over target - i accept all the cost variations and overruns". It stays reported against the target as an accepted overrun, and the savings below remain worth trying.

Main cost drivers (CCK-CAL-001, section 11): the per-channel parts, which are bought eight times (cell holder with four-wire contacts, charger module, channel board, load MOSFET and op-amp loop, thermistor: about \$66 for the eight channels), the mains adapter (\$15), the steel tray with liner (\$12) and the controller board (\$10). The parts added for construction (lines 19 to 21) cost \$8.40.

Savings worth trying, should indicative prices rise above the target:

- Leave the matched-group bins out of the BOM until TRL 4 purchasing (decided 2026-10-02). The fan finger guards (line 10, \$1.20 for two) and the silicone for the wire slots (line 16, \$4.00) decided on 2026-10-02 are now in the estimate; they are the reason it is over the target (accepted by Amish on 2026-10-03).
- Look for a cheaper cell holder or charger module, since each saving is multiplied by eight.

■ Decisions made

Table 3. Decisions made, with dates and records.

Date	Decision	Decided by	Record
2026-09-25	Keep SwapCell in the pitch and plan a second-life variant; charge and discharge attended only (rest stage may be left); hardware watchdog and undervoltage comparators; grading thresholds, 14-day rest and ± 1 % group tolerance as starting values; TP5100, INA226 and ESP32-S3 class modules; eight channels with a linear load; LFP profile later; per-cell CSV mapped to the ReflowEconomy passport (items 1 to 3 and 5 to 9)	Amish: "i accept all your recommendations, go with them across all repos."	CCK-DDR-001, CCK-DDR-002
2026-09-25	Budget raised from \$160 to \$175 (item 4)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-09-25	DC four-wire resistance and a 14-day rest (item 11); firmware and build rules: same-channel cohort self-discharge reading, temperature-gated resistance pulse, per-channel calibration, fan-fault stop, detached-thermistor check (item 13); raise the passport gaps with ReflowEconomy (item 15)	Amish, same instruction	CCK-DDR-001, CCK-DDR-002
2026-10-01	Design for construction: feet and standoffs, 1.5 mm plate on six standoffs, holder fixings and wire slots, folded guard with thumb screws, board standoffs, larger display stand, heatsink and shroud fixings, thermistor C-clips, inlet housing, 5 V converter and channel boards (P1 to P12)	Made under Amish's 2026-09-30 instruction: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The changes themselves were accepted on 2026-10-02 (below)	CCK-DDR-003
2026-10-02	Open item 1: the guard is held down by the two knurled M4 thumb screws for the first prototype, and the TRL 4 containment test is run with the guard held exactly that way; latches are revisited only if the test or users show the screws being left loose	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-003, A1
2026-10-02	Open item 2: each wire slot under the guard is filled round its leads with high-temperature silicone after wiring, closing the guard space again	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-003, A2
2026-10-02	Open item 3: first trial partner, the first candidate to approach (not yet agreed), is a local Repair Café group that already passes laptop batteries to an e-waste or battery collection point, with that collection point as the source of trial cells	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 10
2026-10-02	Open item 4: scheduling rule for the attended day: a charge or discharge starts only if it will finish the same day (12.0 cells per day); no overnight pause	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 12
2026-10-02	Open item 5: R11, the 4.91 kg mass is accepted with the 1.5 mm tray kept, and the prototype is weighed at TRL 4; R5, throughput rests on the	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 14; CCK-CAL-001 section 8

Date	Decision	Decided by	Record
	same-day rule of item 4, and more channels are left for a later version		
2026-10-02	Open item 6: 60 mΩ stays the grade A default; per-model limits are added only for the few cell models most common in the partner's intake, set from DC resistance measured on known-good cells of each model, not from data sheet impedance	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 16
2026-10-02	Open item 7: 18650 cells accepted in the product renders; the model keeps the 21700 envelope for clearance checks	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 8: the matched-group bins are render context only for now; a BOM line is decided at TRL 4 purchasing	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 9: fan finger guards are added to BOM line 10 now, with an estimated price, rather than at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 10: the guard's slotted perforation and 1.0 mm sheet in the renders are accepted as appearance only; the BOM keeps round holes in 0.8 mm sheet	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26
2026-10-02	Open item 11: the second-life pack study (about 334 Wh, 165 W) is sent to the SwapCell and PowerBox projects as a proposal; no further CellCheck action	Amish: "i approve your recommendations for all 555 open decisions."	CCK-DDR-001, item 1; CCK-CAL-001 section 10
2026-10-02	Design for construction accepted: the changes P1 to P12 and their knock-on changes, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	CCK-DDR-003, Tables 1 and 2
2026-10-03	Cost overrun accepted: the estimated cost of USD 177.60 against the USD 175 target (USD 2.60 over; open decision 1, option a)	Amish: "Cost over target - i accept all the cost variations and overruns"	REVIEW.md, session 2026-10-03